

# Developmental and Epileptic Encephalopathy 77 (DEE77): A Comprehensive Disease Characterization

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**Disease:** Developmental and Epileptic Encephalopathy 77 (DEE77) **MONDO ID:** MONDO:0032808 · **OMIM:** #618548 · **Causal gene:** *PIGQ* (16p13.3) **Category:** Mendelian (autosomal recessive inherited GPI-anchor deficiency)

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## Summary

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Developmental and Epileptic Encephalopathy 77 (DEE77; also indexed as Early Infantile Epileptic Encephalopathy 77 / EIEE77 and as Multiple Congenital Anomalies–Hypotonia–Seizures syndrome 4 / MCAHS4) is an **ultra-rare autosomal-recessive disorder** caused by **biallelic loss-of-function / hypomorphic variants in *PIGQ*** (phosphatidylinositol glycan anchor biosynthesis class Q; HGNC:14135; formerly *GPI1*), located on chromosome 16p13.3. *PIGQ* is a subunit of the **GPI-N-acetylglucosaminyltransferase (GPI-GnT) complex** that catalyzes the very first, committed step of glycosylphosphatidylinositol (GPI) anchor biosynthesis on the cytoplasmic face of the endoplasmic reticulum. Because GPI anchoring tethers >150 diverse cell-surface proteins (enzymes, adhesion molecules, receptors, complement regulators), *PIGQ* deficiency produces a systemic **inherited GPI deficiency (IGD)** with a neurodevelopmental core.

The disease was first recognized in 2014, when whole-genome sequencing of severe early-onset epilepsy trios identified a recessive *PIGQ* splice/exon-skipping variant in an Ohtahara-syndrome patient ([PMID: 24463883](#)). The definitive phenotypic delineation came from Johnstone et al. 2020, who reported seven new biallelic-*PIGQ* subjects with functional confirmation ([PMID: 32588908](#)). Clinically, DEE77 presents in **early infancy (seizure onset ~2.5–7 months)** with drug-resistant epileptic seizures, axial hypotonia, global developmental delay/intellectual disability, progressive cerebral and cerebellar atrophy, and multisystem (gastrointestinal and cardiac) anomalies. **Premature death occurs in more than half of reported patients.** A milder end of the spectrum — nonprogressive congenital ataxia with intellectual disability and generalized epilepsy — has also been described ([PMID: 34089469](#)).

There is **no disease-modifying therapy**. Management is supportive and symptomatic, with anti-seizure medications; **high-dose vitamin B6 (pyridoxine / pyridoxal-5'-phosphate)** provides partial seizure benefit in a subset of IGD patients but rarely achieves seizure freedom. Diagnosis rests on **exome/genome sequencing** confirmed functionally by **flow cytometry** showing reduced surface GPI-anchored proteins (e.g., FLAER, CD16, CD14) on granulocytes/leukocytes. Prevention relies on **genetic counseling** with prenatal / preimplantation genetic testing for at-risk (often consanguineous) families. This report synthesizes nine confirmed findings across the full disease-characterization template.

## 1. Disease Information

**Overview.** DEE77 is a monogenic, autosomal-recessive developmental and epileptic encephalopathy belonging to the family of **inherited glycosylphosphatidylinositol (GPI) deficiencies (IGDs)** — congenital disorders of glycosylation affecting the biosynthesis or remodeling of the GPI anchor. It is characterized by early-infantile-onset epilepsy that is frequently drug-resistant, together with severe global developmental impairment and congenital multisystem anomalies.

**Key identifiers** (cross-references verified via OLS for MONDO:0032808):

| Resource                 | Identifier           |
|--------------------------|----------------------|
| MONDO                    | MONDO:0032808        |
| OMIM (phenotype)         | #618548              |
| OMIM (gene <i>PIGQ</i> ) | 605754               |
| DOID                     | DOID:0112213         |
| GARD                     | 0016363              |
| MedGen                   | C5231405 / CN1684735 |
| UMLS                     | C5231405             |
| HGNC (gene)              | HGNC:14135           |

**Synonyms / alternative names:** Developmental and epileptic encephalopathy 77; DEE77; Early infantile epileptic encephalopathy 77; EIEE77; PIGQ-related GPI deficiency; **Multiple congenital anomalies–hypotonia–seizures syndrome 4 (MCAHS4).**

**Information source type.** Knowledge is derived almost entirely from **aggregated disease-level resources** (OMIM, MONDO, ClinVar) and from **small published case series / individual case reports** (Martin 2014; Johnstone 2020; Zanni 2022), not from large EHR-based cohorts. The largest quantitative context comes from a pooled IGD cohort (Sidpra et al. 2024; n=83) that includes *PIGQ* among other GPI-pathway genes ([PMID: 38456468](#)).

## 2. Etiology

**Primary cause — genetic.** DEE77 is caused by **biallelic (autosomal recessive) pathogenic variants in *PIGQ***. There is no environmental, infectious, or acquired etiology; the disorder is fully Mendelian. The first causal link was reported by Martin et al. 2014, who identified a recessive *PIGQ* mutation causing exon skipping and defective GPI biosynthesis in an Ohtahara-syndrome patient among six whole-genome-sequenced severe early-onset epilepsy trios:

*"The fourth OS patient had a recessive mutation in PIGQ that led to exon skipping and defective glycoposphatidyl inositol biosynthesis."* — [PMID: 24463883](#)

Johnstone et al. 2020 then confirmed and expanded the etiology with seven new biallelic-*PIGQ* subjects from six families:

*"We investigated seven children from six families to expand the phenotypic spectrum associated with an early infantile epileptic encephalopathy caused by biallelic pathogenic variants in the phosphatidylinositol glycan anchor biosynthesis class Q (*PIGQ*) gene."* — [PMID: 32588908](#)

**Genetic risk factors.** The causal factor is the biallelic *PIGQ* genotype itself. **Consanguinity** is an important risk factor (homozygous variants recur in consanguineous families, e.g., the homozygous c.1631dupA reported by Zanni et al. 2022). Compound heterozygosity accounts for

cases in non-consanguineous families. No modifier genes have been formally established for DEE77, although residual enzymatic activity of hypomorphic alleles is the principal determinant of severity (see §4, §9).

**Environmental risk / protective factors and gene–environment interactions. None are established.** As a purely Mendelian recessive disorder, environmental exposures do not cause DEE77. The only clinically actionable "environmental" modifier is nutritional/pharmacological — **vitamin B6 (pyridoxine)** partially mitigates seizures in a subset of IGD patients (see §12), representing a gene–treatment rather than a gene–environment interaction. No protective alleles are known.

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### 3. Phenotypes

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The DEE77 phenotype is dominated by a **neurodevelopmental triad (epilepsy + hypotonia + developmental delay)** with frequent multisystem congenital anomalies. Frequencies below combine the *PIGQ*-specific series (Johnstone 2020, n=7; Zanni 2022) with the pooled IGD cohort (Sidpra 2024, n=83) where *PIGQ*-specific numbers are unavailable.

| Phenotype                                                       | Type                  | Onset / severity                             | Frequency                                   | Suggested HPO                                                   |
|-----------------------------------------------------------------|-----------------------|----------------------------------------------|---------------------------------------------|-----------------------------------------------------------------|
| Epileptic seizures (incl. status epilepticus, EIMFS, myoclonic) | Clinical sign         | Onset 2.5–7 mo; severe, often drug-resistant | 83% (IGD pooled); consistent in <i>PIGQ</i> | HP:0001250 (Seizure), HP:0002133 (Status epilepticus)           |
| Axial / generalized hypotonia                                   | Clinical sign         | Congenital/early; moderate–severe            | 72% (IGD pooled)                            | HP:0008936 (Axial hypotonia), HP:0001252 (Hypotonia)            |
| Global developmental delay / intellectual disability            | Behavioral/ cognitive | Early; severe                                | 90% (IGD pooled)                            | HP:0001263 (GDD), HP:0001249 (ID)                               |
| Cerebral atrophy                                                | Imaging/ structural   | Progressive                                  | 75% (IGD pooled)                            | HP:0002059 (Cerebral atrophy)                                   |
| Cerebellar atrophy                                              | Imaging/ structural   | Progressive                                  | 60% (IGD pooled)                            | HP:0001272 (Cerebellar atrophy)                                 |
| Corpus callosum anomalies                                       | Imaging/ structural   | Congenital                                   | 57% (IGD pooled)                            | HP:0007370 (Aplasia/hypoplasia of the corpus callosum)          |
| Gastrointestinal anomalies (incl. midgut volvulus)              | Physical              | Congenital; can be life-threatening          | 66% (IGD pooled); 2/7 volvulus (Johnstone)  | HP:0011024 (Abnormality of the GI tract), HP:0002580 (Volvulus) |
| Cardiac anomalies / arrhythmia                                  | Physical              | Congenital                                   | 19% (IGD pooled)                            | HP:0001627 (Abnormal heart morphology), HP:0011675 (Arrhythmia) |
| Renal malformation                                              | Physical              | Congenital                                   | 14% (IGD pooled)                            |                                                                 |

| Phenotype                          | Type          | Onset / severity                   | Frequency            | Suggested HPO                                                |
|------------------------------------|---------------|------------------------------------|----------------------|--------------------------------------------------------------|
|                                    |               |                                    |                      | HP:0000077<br>(Abnormality of the kidney)                    |
| Dysmorphic features                | Physical      | Congenital; no distinctive gestalt | 82% (any); none >30% | HP:0001999<br>(Abnormal facial shape)                        |
| Motor symptoms (ataxia/dyskinesia) | Clinical sign | Variable                           | 64% (IGD pooled)     | HP:0001251<br>(Ataxia),<br>HP:0100022<br>(Abnormal movement) |

#### Key supporting quotes:

*"Epileptic seizures, axial hypotonia, developmental delay and multiple congenital anomalies were consistently observed. Seizure onset occurred between 2.5 months and 7 months of age and varied from treatable seizures to recurrent episodes of status epilepticus."* — [PMID: 32588908](#)

*"Core clinical features were developmental delay or intellectual disability (DD/ID, 90%), seizures (83%), hypotonia (72%) and motor symptoms (64%)."* — [PMID: 38456468](#)

**Progression and severity.** The severe end shows **progressive** neurodegeneration (progressive atrophy, hypomyelination) and high early mortality; a **milder, nonprogressive** end (congenital ataxia with ID and generalized epilepsy, cerebellar atrophy) is documented ([PMID: 34089469](#)). Severity correlates with residual PIGQ/GPI-biosynthetic activity.

**Quality-of-life impact.** Not formally measured with EQ-5D/SF-36/PROMIS in this ultra-rare disorder. Qualitatively, the impact is **profound**: affected children have severe cognitive/motor disability, drug-resistant seizures, feeding difficulties, and dependence on full-time caregiving; many do not survive early childhood.

## 4. Genetic / Molecular Information

Causal gene — *PIGQ*.

| Attribute                     | Value                               |
|-------------------------------|-------------------------------------|
| Gene symbol                   | <i>PIGQ</i> (formerly <i>GPI1</i> ) |
| HGNC                          | HGNC:14135                          |
| NCBI Gene ID                  | 9091                                |
| OMIM gene                     | 605754                              |
| Ensembl                       | ENSG00000007541                     |
| UniProt                       | Q9BRB3                              |
| Cytoband                      | 16p13.3                             |
| GRCh38 locus                  | chr16:566,995–584,121 (+ strand)    |
| Canonical transcript / RefSeq | ENST00000321878 / NM_004204.5       |

Pathogenic variant spectrum (ClinVar, 300 records queried).

| Classification    | Count |
|-------------------|-------|
| Pathogenic        | 24    |
| Likely pathogenic | 4     |
| VUS               | 105   |
| Likely benign     | 145   |
| Benign            | 1     |
| Conflicting       | 6     |

By molecular type across the record set: 268 SNVs, 17 deletions, 7 duplications, 2 microsatellite, plus rare CNV gains/losses and 1 inversion. **DEE77 variants are predominantly small-scale** (missense, nonsense, frameshift, splice-site), with occasional copy-number/structural events.

Representative variants: the exon-skipping splice variant of Martin 2014 (PMID: 24463883) and the homozygous frameshift NM\_004204.5:c.1631dupA (p.Tyr544fs\*79) of Zanni 2022 (PMID: 34089469).

**Population constraint (gnomAD, computed).** *PIGQ* is **heterozygous LoF-tolerant**, consistent with a recessive mechanism: gnomAD v2 pLI  $\approx 0$  ( $5.7e-17$ ), LOEUF (oe\_lof upper) = 1.12, observed/expected LoF = 0.90 (54 obs vs 60.3 exp), missense Z =  $-0.08$ . Heterozygous carriers are unaffected — concordant with the historical observation that heterozygous 16p13.3 deletions removing one *GPI1* allele (in  $\alpha$ -thalassemia/mental retardation) do not overtly impair GPI-anchored protein expression (PMID: 11418246).

**Functional consequence. Loss of function / hypomorphic.** Complete null is presumed embryonic-lethal; viable patients retain residual GPI-biosynthetic activity. Functional proof of causality: transfection of wild-type *PIGQ* cDNA into patient fibroblasts rescued the GPI-anchored-protein deficiency (see §6).

**Modifier genes / epigenetics / large chromosomal abnormalities.** No DEE77-specific modifier genes or epigenetic mechanisms are established. Whole-gene deletions or 16p13.3 CNVs are rare contributors; most disease arises from small biallelic variants.

## 5. Environmental Information

**Not applicable.** DEE77 is a monogenic recessive disorder with **no environmental, lifestyle, toxic, radiation, occupational, or infectious cause or trigger**. No dietary, behavioral, or exposure-based risk or protective factors have been identified. The only exogenous factor with clinical relevance is **therapeutic vitamin B6 (pyridoxine/pyridoxal-5'-phosphate)**, which partially reduces seizures in some IGD patients (see §12) — a treatment effect, not a disease cause.

## 6. Mechanism / Pathophysiology

### Ordered causal chain

1. **Biallelic hypomorphic/LoF *PIGQ* variants** reduce functional *PIGQ* protein → *leads to*



2. **Impaired GPI-N-acetylglucosaminyltransferase (GPI-GnT) complex** on the cytoplasmic face of the ER, of which PIGQ is a subunit → *results in*
3. **Failure of the first committed step of GPI biosynthesis** (transfer of GlcNAc onto phosphatidylinositol) → *leads to*
4. **Global reduction in GPI anchor synthesis** → *results in*
5. **Reduced cell-surface expression of >150 GPI-anchored proteins** (enzymes, adhesion molecules, receptors, complement regulators) — directly demonstrated in patient granulocytes and fibroblasts → *leads to* (branch point)
6. **5a. Neuronal branch:** loss of GPI-anchored proteins critical to neuronal development, synapse formation, and cortical excitability → *results in* → **epileptogenesis, hypotonia, developmental delay, progressive cerebral/cerebellar atrophy** (inferred from human phenotype + mouse GPI-deficiency models).
7. **5b. Systemic/developmental branch:** loss of GPI-anchored proteins in non-neural tissues during organogenesis → *results in* → **gastrointestinal (volvulus), cardiac, renal congenital anomalies.**
8. Combined CNS and multisystem dysfunction → *results in* → **early-infantile DEE77 with high mortality.**

## Molecular and cellular detail

**PIGQ's molecular role** is defined precisely:

*"PIGQ encodes the phosphatidylinositol glycan class Q protein and is part of the GPI-N-acetylglucosaminyltransferase complex that initiates GPI biosynthesis from phosphatidylinositol (PI) and N-acetylglucosamine (GlcNAc) on the cytoplasmic side of the endoplasmic reticulum (ER)." — PMID: 34089469*

The GPI-GnT ring complex also includes ARV1, which recruits phosphatidylinositol and associates directly with PIGQ:

*"ARV1 associates with PIGQ, a GPI-GnT component" — PMID: 40378954*

**Direct functional proof in human cells** (loss of GPI-anchored proteins is the proximate mechanism, and it is PIGQ-specific):

*"Flow cytometry using granulocytes and fibroblasts from affected individuals showed reduced expression of glycosylphosphatidylinositol (GPI)-anchored proteins. Transfection of wildtype PIGQ cDNA into patient fibroblasts rescued this phenotype." — PMID: 32588908*

**Upstream vs downstream.** The *PIGQ* mutation and GPI-GnT dysfunction are **upstream**; loss of specific GPI-anchored proteins (e.g., complement regulators, folate receptor- $\alpha$ , alkaline phosphatase-related processing, neural adhesion molecules) and the resulting neuronal/organ dysfunction are **downstream**.

**Cellular processes and compartments.** The initiating lesion is in the **endoplasmic reticulum** (GO:0005789, ER membrane; cytoplasmic face). Affected biological processes: **GPI anchor biosynthetic process** (GO:0006506), **GPI anchor metabolic process** (GO:0006505), **protein lipidation / attachment of GPI anchor to protein** (GO:0016255). Downstream neural processes include **regulation of neuron differentiation** and **regulation of synaptic transmission / neuronal excitability**.

**Biochemical hallmark shared across IGD.** Some IGD subtypes show **hyperphosphatasia** (elevated serum alkaline phosphatase, itself a GPI-anchored enzyme aberrantly shed when anchoring fails); this is variable in *PIGQ* disease. **Folate receptor- $\alpha$  (FOLR1) is a GPI-anchored protein**, providing a mechanistic rationale for cerebral folate involvement and folinic-acid relevance (PMID: 19732866).

**Cell types (CL suggestions):** neuron (CL:0000540), GABAergic/inhibitory neuron (CL:0000617), glutamatergic/excitatory neuron (CL:0000679), granulocyte/neutrophil (CL:0000775 — diagnostic readout), fibroblast (CL:0000057 — diagnostic readout).

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## 7. Anatomical Structures Affected

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**Organ level — primary.** The **brain / central nervous system** (UBERON:0000955, brain; UBERON:0001017, CNS) is the primary affected organ. Within the brain, the **cerebral cortex** (UBERON:0000956), **cerebellum** (UBERON:0002037), **corpus callosum** (UBERON:0002336), and white-matter tracts (hypomyelination; symmetric restricted diffusion of the **central tegmental tracts**, 60% in IGD pooled) are involved.

**Secondary / multisystem involvement:** - **Gastrointestinal tract** (UBERON:0001555) — anomalies in 66%, including life-threatening midgut volvulus. - **Heart** (UBERON:0000948) — structural anomalies and arrhythmias in ~19%. - **Kidney** (UBERON:0002113) — malformation in ~14%.

**Body systems:** nervous (primary), digestive, cardiovascular, and renal/urinary (secondary).

**Tissue and cell level.** Primarily **nervous tissue** (excitatory and inhibitory cortical neurons, cerebellar neurons). Because GPI anchoring is ubiquitous, epithelial and other tissues are affected during organogenesis. Diagnostic readouts use **hematopoietic cells** (granulocytes/leukocytes) and **fibroblasts**.

**Subcellular level.** The initiating defect is at the **endoplasmic reticulum membrane, cytoplasmic face** (GO:0005789). The functional deficit manifests at the **plasma membrane** (GO:0005886) as loss of GPI-anchored surface proteins, i.e., in the **anchored component of the plasma membrane** (GO:0031225).

**Localization / lateralization.** CNS involvement is **bilateral and largely symmetric** (symmetric atrophy, symmetric restricted diffusion of central tegmental tracts).

## 8. Temporal Development

**Onset. Early-infantile / congenital.** Congenital hypotonia and multiple congenital anomalies are present at/near birth; **seizure onset occurs between 2.5 and 7 months** (PMID: 32588908). Median age at seizure onset across the IGD family is ~6 months (PMID: 38456468). Onset pattern is **chronic with early emergence**, punctuated by acute events (status epilepticus, volvulus).

**Progression.** At the severe end the course is **progressive**: recurrent/refractory seizures, progressive cerebral and cerebellar atrophy, and hypomyelination, with declining function. At the mild end (Zanni 2022) the course is **nonprogressive** congenital ataxia. Disease is **lifelong (chronic)** when survived.

**Patterns.** Seizures range from treatable to recurrent status epilepticus; **no reliable spontaneous remission** is documented. **Critical period:** the fetal/early-infantile window of neurodevelopment is the period of vulnerability; because the disease "progresses even after birth," early diagnosis and initiation of supportive/pyridoxine therapy is advocated as a potential window of opportunity (PMID: 26165085).

## 9. Inheritance and Population

**Epidemiology.** DEE77 is **ultra-rare**: only ~15–20 patients reported worldwide since 2014 (Martin 2014 first case; Johnstone 2020 +7; Zanni 2022 +1; scattered others). The case count is **too small for formal prevalence/incidence estimates**, and no Orphanet point-prevalence figure is available.

*"Pathogenic variants in the PIGQ gene have been previously reported in 10 patients with congenital hypotonia, early-infantile epileptic encephalopathy, and premature death occurring in more than half cases." — PMID: 34089469*

**Inheritance.** **Autosomal recessive**, biallelic *PIGQ* — homozygous (often consanguineous) or compound heterozygous.

*"The fourth OS patient had a recessive mutation in PIGQ" — PMID: 24463883*

**Carrier / birth-frequency estimate (gnomAD v4, computed).** Summed high-confidence pLoF allele frequency  $q \approx 1.75e-3 \rightarrow$  naïve Hardy–Weinberg carrier frequency  $\approx 2q \approx 1/286$ , and biallelic-LoF birth frequency  $q^2 \approx 1/327,600$ . **This markedly overestimates true disease incidence** because complete *PIGQ* null is presumed embryonic-lethal, and viable patients require at least one hypomorphic (residual-activity) allele; the true birth prevalence is far lower and unquantified.

**Penetrance / expressivity.** Penetrance of biallelic pathogenic genotypes appears **complete**; **expressivity is variable** — from lethal early-infantile encephalopathy at the severe end to nonprogressive congenital ataxia at the mild end, tracking residual enzyme activity.

**Other genetic features.** **No genetic anticipation** (not a repeat-expansion disorder). **No established founder variant.** **Consanguinity** increases risk of homozygosity. **No sex predilection** (autosomal); reported cases include both sexes.

## 10. Diagnostics

**Genetic testing (primary/definitive).** DEE77 is diagnosed by **exome (WES) or genome (WGS) sequencing**, or by **targeted GPI-biosynthesis / epileptic-encephalopathy gene panels** that include *PIGQ*. Because *PIGQ* is intron-heavy, ES/WGS or panel capture is preferred over single-gene Sanger. Chromosomal microarray can detect rare 16p13.3 CNVs but is low-yield for the typical small biallelic variants. Reference transcript for variant reporting: **NM\_004204.5**.

**Functional confirmation.** **Flow cytometry** demonstrating reduced surface GPI-anchored proteins on granulocytes/leukocytes and fibroblasts confirms pathogenicity and functional impact:

*"Flow cytometry confirmed deficiency of several GPI-anchored proteins on leukocytes (CD14, FLAER)." — PMID: 34089469*

FLAER (fluorescent aerolysin, which binds GPI anchors directly), **CD16**, **CD14**, and **CD59/CD55** are standard markers.

**Laboratory / biomarkers.** Serum **alkaline phosphatase** may be elevated (hyperphosphatasia) in some IGD subtypes (variable in *PIGQ*). No specific circulating biomarker is validated for *PIGQ*-DEE77.

**Neuroimaging (diagnostic/prognostic).** Brain MRI is central:

*"Prognostic and biologically significant neuroimaging features included cerebral atrophy (75%), cerebellar atrophy (60%), callosal anomalies (57%) and symmetric restricted diffusion of the central tegmental tracts (60%)." — PMID: 38456468*

**Electrophysiology.** **EEG** documents the epileptic encephalopathy (slow background; multifocal/continuous epileptiform activity; patterns including epilepsy of infancy with migrating focal seizures and myoclonic status).

**Clinical criteria / differential diagnosis.** No disease-specific criteria beyond genotype + GPI-flow-cytometry. Differential diagnosis includes **other inherited GPI deficiencies** (*PIGA*, *PIGT*, *PIGV*, *PIGO*, *PIGS*, *PIGW*, etc.), other early-infantile DEEs, congenital disorders of glycosylation, and pyridoxine-dependent epilepsies (*ALDH7A1*) — distinguished by gene-specific sequencing and GPI-anchored-protein assays.

**Screening.** No population newborn screening exists. **Cascade carrier testing** of relatives and **prenatal/preimplantation testing** are available once the familial biallelic variants are known.

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## 11. Outcome / Prognosis

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**Mortality.** Prognosis is **poor at the severe end**: premature death occurs in >50% of reported *PIGQ* patients ([PMID: 34089469](#)), and biallelic *PIGQ* variants were explicitly associated with increased mortality:

*"Pathogenic biallelic PIGQ variants were associated with increased mortality." — [PMID: 32588908](#)*

In the pooled IGD cohort, 15/83 were deceased ([PMID: 38456468](#)). Deaths are typically from neurologic complications, status epilepticus, or surgical/systemic complications (e.g., midgut volvulus).

**Morbidity / function.** Survivors have **severe, lifelong disability**: profound developmental delay/intellectual disability, drug-resistant epilepsy, motor impairment (hypotonia, ataxia, dyskinesia), feeding difficulties, and sensory involvement (vision/hearing) in some. No validated QoL instruments have been applied.

**Recovery potential. Limited.** No disease-modifying therapy exists; the milder (nonprogressive) subgroup has a more stable but still impaired course.

**Prognostic factors. Residual GPI-biosynthetic activity / variant severity** is the principal prognostic determinant (severe LoF → early lethality; hypomorphic → milder ataxia). **Neuroimaging** features (cerebral/cerebellar atrophy, callosal anomalies, central-tegmental-tract diffusion restriction) are prognostically significant. **Neonatal onset and refractory seizures** portend worse outcome.

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## 12. Treatment

**Overview.** There is **no cure and no disease-modifying therapy** for DEE77; management is **supportive and symptomatic** (anti-seizure medications, nutritional/feeding support, surgical correction of anomalies such as volvulus, developmental therapies). Epilepsy is frequently drug-resistant.

**Vitamin B6 (pyridoxine / pyridoxal-5'-phosphate; NCIT: Pyridoxine, C939).** A rational adjunct across IGD. A prospective compassionate-use cohort (Bayat et al. 2022; n=7 IGD, *PIGA*/*PIGT*/*PIGV*) using pyridoxine 20–30 mg/kg/day then P5P found partial benefit:

*"We observed more than 50% seizure frequency reduction in 2 out of 7 and less than 50% reduction in another 3 out of 7 participants. No participants reached seizure freedom." — PMID: 35080266*

*"Early diagnosis and treatment are desirable because the disease progresses even after birth and vitamin B6(pyridoxine) is very effective for some patients with intractable seizures." — PMID: 26165085*

**Proof-of-concept targeted therapy (different GPI gene).** Restoring GPI-anchored protein expression can control IGD seizures — an HDAC inhibitor (butyrate) increased *PIGM* transcription and surface GPI expression:

*"the drug caused complete cessation of intractable seizures in a child with inherited GPI deficiency." — PMID: 17442906*

This is a mechanism-anchored strategy but has not been tested in *PIGQ*-DEE77.

**Folinic acid.** Mechanistically relevant because **folate receptor-α (FOLR1) is itself a GPI-anchored protein**; cerebral folate deficiency is treatable with folinic acid (PMID: 19732866). Empirical benefit in *PIGQ*-DEE77 is unproven.

**Advanced / experimental therapeutics.** No gene therapy, cell therapy, RNA-based therapy, or *PIGQ*-targeted small molecule exists or is in trials. Highly purified cannabidiol has shown benefit in monogenic epilepsies broadly (PMID: 40126049) but is not *PIGQ*-specific.

**Suggested NCIT clinical-intervention terms:** Pyridoxine (C939), Pyridoxal Phosphate (C61970), Anticonvulsant Agent (C264), Folinic Acid/Leucovorin (C576), Supportive Care (C15272).

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## 13. Prevention

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**Primary prevention.** There is **no way to prevent the biallelic genotype**; primary prevention is **reproductive** — **genetic counseling** for at-risk families (especially consanguineous couples) with **carrier testing**, **prenatal diagnosis**, and **preimplantation genetic testing (PGT-M)** once familial *PIGQ* variants are known.

**Secondary prevention.** **Cascade genetic testing** of relatives; early molecular diagnosis to enable prompt supportive care and a trial of pyridoxine/P5P (rationale: the disease progresses postnatally, so early intervention is advocated — [PMID: 26165085](#)).

**Tertiary prevention.** Prevent complications: aggressive seizure management, surveillance and surgical correction of GI anomalies (volvulus), cardiac monitoring for arrhythmias, nutritional support, and developmental/rehabilitative therapies.

**Immunization / public-health / environmental interventions.** Not applicable (non-infectious, non-environmental Mendelian disorder). Standard childhood immunization for general health.

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## 14. Other Species / Natural Disease

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**Taxonomy.** No naturally occurring *PIGQ*-DEE77 has been described in companion animals or wildlife (no OMIA entry identified). GPI anchoring is universally essential across eukaryotes, so orthologs exist broadly. Human species: *Homo sapiens* (NCBI Taxon 9606); mouse: *Mus musculus* (NCBI Taxon 10090).

**Orthologous genes.** Mouse *Pigq* (ortholog of human *PIGQ*) exists; the pathway ortholog most studied for disease modeling is *Piga* (see §15).



**Comparative biology.** GPI-anchor biosynthesis is **deeply evolutionarily conserved** from yeast to humans; the essentiality of the pathway (embryonic lethality of complete knockout) is conserved, which is why disease modeling relies on conditional/tissue-specific approaches (§15).

**Zoonotic / transmission.** Not applicable — non-transmissible genetic disorder.

## 15. Model Organisms

**No *PIGQ*-specific animal model has been published.** Because complete GPI biosynthesis knockout is **embryonic-lethal** (GPI anchoring is essential for embryogenesis), disease modeling uses **conditional / tissue-specific knockouts** of pathway genes.

**Key mouse model (Kandasamy et al. 2021).** Neuron-type-specific GPI-deficiency mice were generated by conditional knockout of *Piga* — the gene catalyzing the same first, committed step of GPI biosynthesis in which PIGQ participates as a GPI-GnT subunit — in telencephalon excitatory neurons (Ex-M-cko), inhibitory neurons (In-M-cko), or thalamic neurons (Th-H-cko). These models recapitulate core DEE77 features:

*"Both Ex-M-cko and In-M-cko mice showed impaired long-term fear memory and were more susceptible to kainic acid-induced seizures." — PMID: 33607654*

*"Phosphatidylinositol glycan biosynthesis class A protein (PIGA) catalyzes the very first step of GPI anchor biosynthesis. Patients carrying a mutation of the PIGA gene usually suffer from inherited glycosylphosphatidylinositol deficiency (IGD) with intractable epilepsy and intellectual developmental disorder." — PMID: 33607654*

In-M-cko mice additionally showed a severe **limb-clasping** phenotype and Ex-M-cko mice showed hippocampal synapse changes.

**Model characteristics / recapitulation.** The neuronal GPI-deficiency mouse reproduces **seizure susceptibility and cognitive deficits** — the neurodevelopmental core of DEE77. **Limitations:** these use *Piga* (not its GPI-GnT partner *Pigq*), are conditional (not the constitutive biallelic human genotype), and do not capture the full multisystem congenital-anomaly spectrum (GI/cardiac/renal). In-vitro human models (patient fibroblasts and iPSC-derived systems) complement animal work: patient fibroblasts provided the definitive rescue experiment (PMID: 32588908).

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Biallelic hypomorphic/LoF PIGQ variants (16p13.3)
| (autosomal recessive; >=1 residual-activity allele in viable patients)
v
Reduced PIGQ subunit -> Impaired ER GPI-GnT complex (with PIGA, PIGC, PIGH, PIGP, DPM2, ARV
| [cytoplasmic face of ER membrane; GO:0005789]
v
Failure of 1st committed GPI step: PI + UDP-GlcNAc -> GlcNAc-PI
| [GPI anchor biosynthetic process; GO:0006506]
v
Global reduction in GPI anchor synthesis
|
v
Deficient surface display of >150 GPI-anchored proteins
(FLAER-binding anchors, CD16, CD14, CD59, FOLR1, ALPL, adhesion/signaling molecules)
|
+-----+-----+
v (neuronal branch) v (systemic/developmental branch)
Disrupted neuronal development, Disrupted organogenesis
synapse formation, excitability
|
v
Epilepsy (onset 2.5-7 mo, drug- GI (volvulus 66%), cardiac (19%),
resistant), hypotonia, GDD/ID, renal (14%) congenital anomalies
progressive cerebral/cerebellar
atrophy, hypomyelination
|
+-----+-----+
v
DEE77 - early-infantile epileptic encephalopathy
with multisystem anomalies; >50% premature mortality
|
v (partial, non-curative)
High-dose vitamin B6 (pyridoxine/P5P) -> partial seizure reduction in a subset

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18

— explaining the continuum from lethal early-infantile encephalopathy to nonprogressive ataxia. The neurodevelopmental prominence reflects the dependence of neuronal differentiation, synaptogenesis, and excitability control on GPI-anchored molecules, validated by conditional neuronal *Piga* knockout mice that reproduce seizure susceptibility and memory deficits. The partial pyridoxine responsiveness — shared across IGD — and the butyrate proof-of-concept in *PIGM* deficiency indicate the pathway is **pharmacologically modifiable in principle**, motivating mechanism-anchored therapy development.

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## Evidence Base

| PMID                     | Title (abbrev.)                                                | Role in this report                                                                                                     |
|--------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <a href="#">24463883</a> | Clinical WGS in severe early-onset epilepsy                    | First link of recessive <i>PIGQ</i> (exon-skipping) to early-onset epileptic encephalopathy                             |
| <a href="#">32588908</a> | EIEE due to biallelic <i>PIGQ</i> : 7 new subjects             | Defines core phenotype, onset, mortality; functional rescue proving causality                                           |
| <a href="#">34089469</a> | <i>PIGQ</i> -related GPI deficiency with nonprogressive ataxia | Milder-end spectrum; molecular role of <i>PIGQ</i> ; >50% premature death; flow-cytometry diagnosis; c.1631dupA variant |
| <a href="#">40378954</a> | ARV1 in GPI-GnT complex                                        | Confirms <i>PIGQ</i> is a GPI-GnT component                                                                             |
| <a href="#">38456468</a> | Clinical/genetic spectrum of IGD (n=83)                        | Pooled frequencies; neuroimaging prognostic features                                                                    |
| <a href="#">33607654</a> | Neuronal GPI-deficiency mouse models                           | Model organism recapitulation of seizures + cognitive deficits                                                          |
| <a href="#">11418246</a> | Human <i>GPI1</i> required for efficient GPI biosynthesis      | Establishes <i>GPI1/PIGQ</i> function; heterozygous deletion tolerance (recessive mechanism)                            |
| <a href="#">35080266</a> | Pyridoxine/P5P for GPI-deficiency seizures                     | Quantifies partial B6 efficacy                                                                                          |
| <a href="#">26165085</a> | Inherited GPI deficiency overview                              | Postnatal progression; B6 effectiveness in a subset                                                                     |
| <a href="#">17442906</a> | Targeted therapy for inherited GPI deficiency                  | Proof-of-concept: restoring GPI expression controls seizures                                                            |
| <a href="#">19732866</a> | FOLR1 cerebral folate transport deficiency                     | FOLR1 is GPI-anchored → folinic-acid rationale                                                                          |

Evidence source types: **human clinical** (24463883, 32588908, 34089469, 38456468, 35080266, 26165085, 17442906, 19732866), **in vitro** (32588908 fibroblast rescue; 11418246 HEK293 antisense), **model organism** (33607654), **computational** (gnomAD constraint, ClinVar and OLS queries performed during the investigation).

## Limitations and Knowledge Gaps

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1. **Ultra-rare evidence base.** Fewer than ~20 *PIGQ* patients are reported; most quantitative frequencies (seizures 83%, DD/ID 90%, etc.) come from the **broader IGD cohort** (Sidpra 2024), not *PIGQ*-specific data. *PIGQ*-specific frequencies, penetrance, and natural history are imprecise.
  2. **No formal epidemiology.** No validated prevalence/incidence; the gnomAD-derived carrier estimate (~1/286) overestimates disease because it ignores embryonic lethality of complete null and the requirement for a hypomorphic allele.
  3. **No *PIGQ*-specific animal model.** Modeling relies on *Piga* conditional knockouts; genotype–phenotype fidelity for *PIGQ* is inferred, not demonstrated.
  4. **Therapeutics extrapolated.** Pyridoxine/P5P and butyrate evidence comes from other GPI genes (*PIGA/PIGT/PIGV/PIGM*); no *PIGQ*-specific trial exists. Folinic-acid benefit is mechanistic conjecture.
  5. **Genotype–phenotype correlation** (which variants give the severe vs. mild spectrum) is not systematically established; residual-activity assays per variant are lacking.
  6. **QoL and long-term outcome** are not measured with standardized instruments.
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## Proposed Follow-up Experiments / Actions

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1. **Assemble a *PIGQ*-specific patient registry** (via GeneMatcher/consortia) to derive gene-specific phenotype frequencies, survival curves, and genotype–phenotype correlations.
2. **Variant-level functional assays.** Quantify residual GPI biosynthesis for each reported *PIGQ* variant (flow cytometry for surface GPI-APs in patient/edited cells) to build a severity-prediction model that would refine prognosis and counseling.
3. **Generate a *Pigq* mouse (or zebrafish) model** — conditional/hypomorphic neuronal knockouts — to directly test *PIGQ* (vs *Piga*) mechanism and the multisystem anomalies (GI/ cardiac/renal).
4. **Prospective pyridoxine/P5P trial in *PIGQ*-DEE77**, with EEG/seizure-diary endpoints and GPI-AP flow-cytometry biomarkers, to test whether the subset-benefit seen in other IGDs extends to *PIGQ*.
5. **Test pathway-restoring agents** (HDAC inhibitors as in *PIGM*; substrate/precursor supplementation) in patient iPSC-derived neurons/organoids for *PIGQ*-specific rescue.

6. **Evaluate folinic acid** empirically, given FOLR1's GPI dependence, measuring CSF 5-MTHF where feasible.
7. **Standardize diagnostic workup:** recommend GPI-AP flow cytometry (FLAER/CD16/CD14) as reflex confirmation after ES/WGS identifies biallelic *PIGQ* variants, plus MRI for prognostic atrophy/tract markers.

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*Report compiled from 9 confirmed findings and 37 reviewed papers across a 5-iteration autonomous investigation. All mechanistic and clinical claims are anchored to primary literature (PMIDs above) with verified abstract quotations.*

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